

Sheng'en Li

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| Kunshan, China

EDUCATION

Duke Kunshan University & Duke University

Expected Dec. 2026

B.S. in Applied Mathematics (dual-degree program)

GPA: 3.80/4.00

- **Honors:** Dean's List with Distinction (Fall 2023, Spring 2024); Summer Research Scholar (2024, 2025).

RESEARCH INTERESTS

Reliable and trustworthy machine learning; causal and counterfactual inference for adaptive decision systems; graph and temporal learning; scientific machine learning, model auditing, and numerical verification.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed publications

- [1] **Sheng'en Li**, Dongmian Zou. "COPF: An Online Framework for Deployment-Stable Counterfactual Fairness in Evolving Graphs." *International Conference on Machine Learning (ICML)*, 2026. **First author.** [\[paper\]](#)
- [2] Shoju Wang*, Yuchen Song*, **Sheng'en Li***. "Enhancing Fairness in Autoencoders for Node-Level Graph Anomaly Detection." *European Conference on Artificial Intelligence (ECAI)*, 2025. **Co-first author.** [\[paper\]](#)

Major manuscripts

- [3] **Sheng'en Li*** et al. "COMET: Counterfactual Moment Auditing under Weak Overlap and Dependent Feedback." **First author.**
- [4] Juntang Wang*, **Sheng'en Li***, Yihan Wang*, Jiayu Gao*, Duanduan Wang*, Dongmian Zou, Shixin Xu. "Auditing and Refining Partial-Physics Models: Scaffold Adequacy, Local Identifiability and ECG-Based Electrolyte Inference."

Selected collaborative manuscripts – co-first author on each

- [5] "Pambda: Privacy Alignment with Calibration." Under review at KDD 2027.
- [6] "LambdaLens: Auditing a Representation-Changing Hyperparameter in Graph Neighbor Embedding." Under review at KDD 2027.
- [7] "MHeatmap: Two-Walk Laplacian Reordering and Mosaic Layout for Rectangular Matrix Visualization." Under review at KDD 2027.
- [8] "DriftNeRF: Cross-Ray Density Drifting for Sparse-View Neural Radiance Fields." Under review at KDD 2027.
- [9] "FToken: Factors as Tokens for Dual-Axis Cross-Sectional Return Prediction." Under review at KDD 2027.

*Equal contribution. Author lists are abbreviated for selected collaborative manuscripts.

SELECTED RESEARCH EXPERIENCE

Counterfactual Learning and Auditing for Adaptive Decision Systems

Jan. 2025 – Sep. 2026

Independent research, Duke Kunshan University / Duke University; advised by Prof. Dongmian Zou and Prof. Fan Wei

- Developed **COPF**, an online framework for fair evolving-graph recommendation, integrating exploration and propensity logging, graph-aware doubly robust counterfactual estimation, residual outcome-indistinguishability auditing, and primal–dual fairness control.
- Extended the framework in **COMET** to certify counterfactual moments under weak overlap, heavy-tailed outcomes, graph–time dependence, and distribution drift, using sample-split subgroup discovery and certificate-guided policy repair.
- **Outcomes:** COPF accepted to **ICML 2026** (first author); COMET in preparation (first author).

Audit–Refine Partial-Physics Models for ECG-Based Electrolyte Inference		Mar. 2026 – Aug. 2026
<i>Collaborative research, Duke Kunshan University; advised by Prof. Shixin Xu and Prof. Dongmian Zou</i>		
- Co-developed a model-auditing pipeline separating scaffold adequacy from inverse identifiability through projective alignment, rank-matched null models, residual–concentration analysis, and tangent/Fisher diagnostics. - Built and validated analyte-specific MIMIC-IV-ECG pipelines; implemented PPIDM-II components and evaluated potassium-specific reduced-rank refinements on held-out data. Outcome: Journal manuscript in preparation on auditing and refining partial-physics models for ECG-based electrolyte inference.		
Verified Taylor-Model Flowpipes in PyTorch		May 2026 – Present
<i>Research collaboration with Prof. Huan Zhang’s group, University of Illinois Urbana-Champaign</i>		
- Developing a differentiable Taylor-model flowpipe engine in PyTorch and a comparison suite against Flow*, covering interval/polynomial arithmetic, Picard propagation, remainder validation, and CPU/GPU consistency tests. - Identified repeated materialization of shared remainders across nonlinear paths as a source of long-horizon enclosure growth; developed step-level diagnostics, regression tests, and controlled tightness/runtime comparisons.		
Multivariate Time-Series Anomaly Detection via SPD Geometry		Jun. 2025 – Jun. 2026
<i>Independent research, Duke Kunshan University; advised by Prof. Dongmian Zou</i>		
Developed a normalizing-flow approach over symmetric-positive-definite representations of multivariate sequences and evaluated anomaly detection across heterogeneous datasets and distribution shifts.		
Fair Graph Anomaly Detection		Jul. 2024 – May 2025
<i>Summer Research Scholar, Duke Kunshan University; advised by Prof. Dongmian Zou</i>		
- Developed a fairness-aware graph autoencoder for node-level anomaly detection, incorporating a counterfactual fairness objective based on causal interventions. - Evaluated the fairness–utility trade-off through multi-dataset experiments and ablation studies. Outcome: Co-first-author paper accepted to ECAI 2025.		
INDUSTRY EXPERIENCE		
Quantitative Developer Intern		Jul. 2026 – Present
<i>Shangchen Private Fund, Shanghai, China</i>		
- Built and validated Python/SQL research-data pipelines for financial filings, report metadata, and point-in-time records; supported factor/liquidity research through consistent definitions and reproducible validation. - Developed internal LLM-agent workflows for financial-document retrieval and database-backed question answering, emphasizing evidence traceability, temporal and metric-definition constraints, and repeatable outputs.		
LEADERSHIP AND SERVICE		
Resident Assistant, Duke Kunshan University		Aug. 2024 – May 2025
Volunteer Mathematics Instructor, Tongren, China		Jul. 2024
Group technical support: GPU/server maintenance and junior-student mentoring		2025 – Present
TECHNICAL SKILLS		
Programming: Python, C++, SQL, MATLAB; working knowledge of Java and JavaScript.		
Frameworks and tools: PyTorch, NumPy, pandas, scikit-learn, Git/GitHub, Linux, LaTeX, pytest, MySQL/SQLite.		
Methods: causal inference and off-policy evaluation, graph neural networks, temporal modeling, diffusion and normalizing-flow models, numerical verification, experiment design and reproducibility.		